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Harmonising research data to support mental health services decision making and care planning: development of a data harmonisation platform in regional Australia

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Abstract

Background This paper outlines the development of a new, collaborative data harmonisation initiative within the Hunter New England (HNE) region of New South Wales in Australia. The initiative aims to address the issue of fragmented population mental health data across health services, research institutions and government agencies through development of a standardised platform that facilitates sharing and harmonisation of data across diverse datasets. By enhancing the region's data integration capabilities, incorporating both new and retrospective data, the platform will support emerging collaborative research initiatives and improve decision-making efficiency and effectiveness in mental health service planning. Data harmonisation is essential for mental health research because it enables large-scale, high-quality studies that can uncover patterns, causes, and solutions that would otherwise remain hidden in fragmented datasets. Mental health conditions are highly variable in their presentation, progression, and response to treatment, making it crucial for researchers to analyse diverse data sources—spanning clinical records, epidemiological studies, social determinants, genetics, and real-world patient experiences. Furthermore, population data suggests that fewer than 50% of people experiencing mental health difficulties seek treatment. Without harmonisation, inconsistencies in data collection and classification limit the ability to compare studies, replicate findings, or generate insights that are applicable across different populations and healthcare settings. Platforms supporting harmonisation of data can be enhanced with real-time data queries and visualisation, providing a unique way to understand, monitor and address population health priorities that have national significance.

Method Key stakeholders contributed through formation of a steering committee with clinical oversight and governance, and by providing feedback during development, resulting in a preliminary prototype data dashboard, focused on mental health data. Key aims of the platform were ease and suitability for use by various end-user groups.

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Results/Discussion The first version of the prototype dashboard is being developed in the Power BI platform, focusing on the mental health domain. It is hoped that the platform will be piloted in the next phase of our project to inform scalability and potential for broader rollout within other regional and metropolitan areas of Australia.

Keywords Mental health, Data harmonisation, Heterogeneity

Background

Timely access to health data is crucial for researchers, clinicians and policy makers to implement effective decision making, service planning and implementation of health initiatives at a local, regional and national level. Data sharing has long been recognised as pivotal to the advancement of scientific progress and discovery [8, 20] with numerous benefits, including improving the replicability of research findings; supporting collaboration and exploration of new research questions; and maximising the investment made in existing large-scale trials, cohort studies and health initiatives [3, 8, 15]. These benefits translate to more efficient exchange, synthesis and reporting of data to drive new research and collaborations, and inform policy and practice [10, 31, 32]. Effective reuse of data also upholds an ethical responsibility to minimise the burden of data collection to individuals, particularly those from vulnerable populations.

In Australia, where mental health services are highly decentralised and vary significantly between states, data harmonisation is essential to achieving a truly national approach to mental health care and research. Mental health data is highly fragmented across primary care (e.g. general practitioners), specialist services (e.g. psychiatrists, psychologists), hospitals, community services, and crisis intervention programs, and is deeply intertwined with non-health sectors, such as education, employment, housing, and the justice system. Without harmonised data across these domains, researchers and policymakers are left with incomplete pictures, limiting their ability to develop targeted interventions, track patient journeys, and assess the effectiveness of treatments and policies at scale. This hampers data-driven decision-making in mental health and the development and mobilisation of a learning health system where data are continuously collected, analysed, and applied to improve healthcare in real time.

Given advances in large-scale data infrastructure (e.g., Australian Research Data Commons [4, 7, 14]) along with newer integrative methodologies such as data harmonisation procedures [2, 9] and individual participant data analysis [11, 20] data can be aggregated and analysed both within and across multiple contexts. Data harmonisation refers to a collection of processes that allows data from independent sources to be transformed and combined, making data more comparable and, where feasible, producing aggregated datasets that can be further explored to address new or more extensive research

questions [9, 14]. By enabling access to larger, more heterogeneous datasets that better represent variance within the population of interest, researchers and decision makers can aggregate, filter and process the data in different ways according to the purpose and research question.

To date, there have been numerous large-scale data initiatives involving the integration of datasets at a national or international level. Examples include HeSANDA (Australian Research Data Commons [4]), a digital catalogue of metadata from research studies willing to share data; the *Intensive Longitudinal Health Behavior Network* in the US [10]; the MINDMAP project [7]; and the Maelstrom platform [14], which provides a catalogue of study metadata spanning numerous international research networks for easier identification of relevant datasets. The strength of these initiatives is that they are often well-funded, and as the datasets grow in size and diversity, they attract more data contributors and researchers over time – thus facilitating more complex analyses and research questions. These larger initiatives tend to have a national or international focus, though they vary in terms of the degree of statistical harmonisation applied to the underlying datasets.

Conversely, other research initiatives have utilised data harmonisation procedures as a methodological tool for enabling more advanced statistical analysis and exploration within a more tightly scoped context. These projects are usually focused on a specific cohort and rigorously controlled, ensuring methodological consistency and a high degree of data uniformity. One example is the NEW1000 study [17], a collaboration between the University of Newcastle, Hunter Medical Research Institute and the Hunter New England Health Local Health District. The NEW1000 project applies data harmonisation techniques to multimodal data (questionnaire, physical assessment, phenotype data and biological samples), thereby enabling a diverse team of researchers to better understand the impact of the first 1000 days of a child's life across various domains including health, social development and wellbeing.

Many current data harmonization initiatives store data on a central server-based system, however, since the COVID-19 pandemic healthcare systems across the world have been moving away from a centralised system to more decentralised (cloud based) peer to peer network to address some of the user privacy, confidentiality and trust in the system issues. The main advantage is that users, are issued as a Digital Wallet with all personal

health record (PHR) securely stored in “Health Passport” (cloud based system) [16], offering users greater control over their personal health information and capacity to share PHR record with potential data harmonisation projects.

A recent systematic review of ethics and legislation of global patient co-owned cloud (GPOC) systems for personal health data has shown that only two-thirds of countries have data protection and privacy laws in place, including Australia and the global PHR producers are dominated by the US, with EPIC Systems Corporation (EPIC) and Oracle Cerner, health care systems having the largest share of global use [24]. This review also highlighted the ethical challenges around the integration of AI into healthcare systems, particularly around privacy and data security, and potential safeguards for ethical AI use within the GPOC should be included [24]. In Australia, we are in the early stages of adopting digital healthcare systems (e.g., EPIC), and it will be interesting to see what impact the introduction of new healthcare systems (i.e., single digital patient record) has on data harmonisation initiatives and whether these systems are more adept at addressing ethical concerns around data/user privacy.

To our knowledge, there are few cases where data harmonisation has been applied effectively to support health systems planning and decision making more broadly, at a local or regional level, and none in mental health. While some high-level aggregate mental health data is available, such as ambulance and hospital presentations via state-based suicide surveillance systems, these tend to capture only a fragment of the most serious presentations. In fact, population data shows that fewer than 50% of individuals with mental health concerns seek treatment via traditional service pathways (Australian Institute of Health and Welfare [5]). Conversely, in a local health context, large amounts of individual patient data are routinely collected via patient-facing services as well as through community-based programs and university-led clinical trials. Data harmonisation initiatives in this context may therefore incorporate data from research studies, administrative datasets or clinical data alongside demographic or other health services or population-level data [30, 32] to form a more complete picture of functioning at the population level. Harmonising across these datasets has numerous potential benefits for multiple stakeholders, including the research community, clinicians, jurisdictional public health units and policy makers. However, despite calls for reuse and sharing of clinical and health services data, much of the data that is currently collected remains fragmented, siloed and under-utilised [12, 28].

There is a need for new regional data harmonisation platforms that allow existing health data to be used to greater effect for health systems planning, policy, research and service delivery at a local level. With new

research findings taking up to 17 years to make their way into clinical practice [33, 40], new data initiatives are critical for bridging existing knowledge translation gaps and enabling the timely application of health knowledge and behaviour change research findings within clinical practice and service delivery. In Australia, health care services, community-based services and primary health networks are funded by the federal government but delivered by the State. The location of services is determined by the population growth and needs of the community; hence services tend to be in metropolitan, rural and regional communities. Thus, data initiatives should have a local or regional lens but also support the aggregation of health service and research data with larger-scale datasets generated through population health initiatives and epidemiological studies.

Unsurprisingly, there are significant challenges to consider prior to launching any such initiative. A future challenge following the launch of the platform will be the process of educating stakeholders and embedding data sharing into routine practice. Prior research suggests that while many researchers are supportive of data sharing at a conceptual level, there is a substantial gap in terms of how many actually share their data [36]. On one hand, data custodians may be reluctant to share due to ethical constraints or queries around data ownership, privacy and security; while limited resources and technical and methodological challenges can further impede sharing [20, 36]. These challenges may be amplified in the context of health service data, where staff and administrative resources are typically limited, and data sharing may be constrained by strict organisational regulations governing security, privacy and confidentiality. There are also considerations around data sovereignty and governance, particularly with regard to data collected from, and about, Australia's First Nations peoples [23, 38]. Furthermore, uncertainty around whether the sharing of data across jurisdictional units falls outside of its original intent or requires a waiver of patient consent can also limit efforts to coordinate and share data.

The data harmonisation project aimed to inform the development of a prototype. This paper describes the process of a data harmonisation initiative that seeks to address some of these challenges, initially scoped to the Hunter New England (HNE) region of New South Wales, Australia. The Hunter New England (HNE) region serves as a microcosm of the rest of Australia due to its diverse geography, population, and healthcare challenges, which reflect national trends (Australian Bureau of Statistics [6]). The region encompasses major urban centres, regional hubs, rural towns, and remote communities, mirroring Australia's overall distribution of population density and access to services. It has a mix of industries, including mining, agriculture, health, education, and

tourism, representing key sectors in the national economy. HNE is also home to Aboriginal communities, aging populations, and socioeconomically diverse groups, reflecting Australia's broader demographic composition and health inequities. The region experiences similar public health challenges to the rest of the country, including chronic disease, mental health issues, and healthcare access disparities between urban and rural areas. The region has a comprehensive health system that spans the full spectrum of care, from tertiary referral hospitals and specialist services to primary care, community health, and preventive programs. This integration allows for a whole-of-system approach to health improvement and research translation. Furthermore, HNE benefits from a rich level of health service data, including hospital, primary care, and population health datasets, which can support learning health systems, data-driven decision-making, and research innovations that have national relevance. Additionally, the HNE region is home to a dynamic and adapting health and research ecosystem that embraces innovation in models of care and digital infrastructure. For example, HNE will be the first health district in NSW to roll out the new, Government-led Single Digital Patient Record system, an integrated health system designed to streamline the collection and usage of secure clinical records across services.

The size and diversity of the region make HNE an ideal setting for health and medical research, policy trials, and service innovations that can be scaled to benefit the broader Australian population. This paper outlines the key steps from the development of a new

data harmonisation initiative. Key research questions we set out to address included (1) can we effectively build awareness, engagement and 'buy in' to the data harmonisation initiative amongst local researchers, affiliates and health services; (2) what are the required processes and challenges of collating deidentified mental health data from researchers and clinicians within the region; and (3) what elements are important for building a user-friendly data platform that addresses the diverse needs of the key stakeholder groups. Project activities included establishing the project plan and scope through stakeholder mapping; formation of a steering committee and development of a structured data governance and data contribution framework; applying data harmonisation procedures to the collated data; and building a prototype data dashboard.

Methods

Early involvement from representatives of key user groups was considered critical to the success of the project. We initially identified three overlapping core stakeholder groups who all make unique contributions to the multi-directional flow of health data and information within the region (see Fig. 1): *clinicians*, who may use the platform for health service planning, informing service delivery, and identifying gaps and areas of unmet need; *researchers*, who may use the platform for secondary data analysis and forming new collaborations; and—reflective of the diversity of the HNE region (Australian Bureau of Statistics [6]), representatives from priority groups including people with lived experience and Aboriginal and Torres Strait Islander peoples, who can provide valuable input around the use of visualisations and terminology, as well as ownership, usage and protection of data (in accordance with the principal researcher/institution). For the current project, we utilised a co-design approach to engage with representatives from each group through the formation of a steering committee and to obtain feedback around the priorities and data requirements for each user group during development.

We adopted an iterative co-design approach guided by regular feedback loops with stakeholders to adjust and refine the platform based on end-user input. Co-design activities included group discussions and review sessions where members were shown early working versions of the platform prototype and asked for feedback on usability, appropriateness, and relevance.

Results

Project planning and stakeholder mapping

Project planning began with a series of initial consumer engagement activities designed to determine feasibility and interest in the initiative. These activities included consultation with clinicians and researchers who are

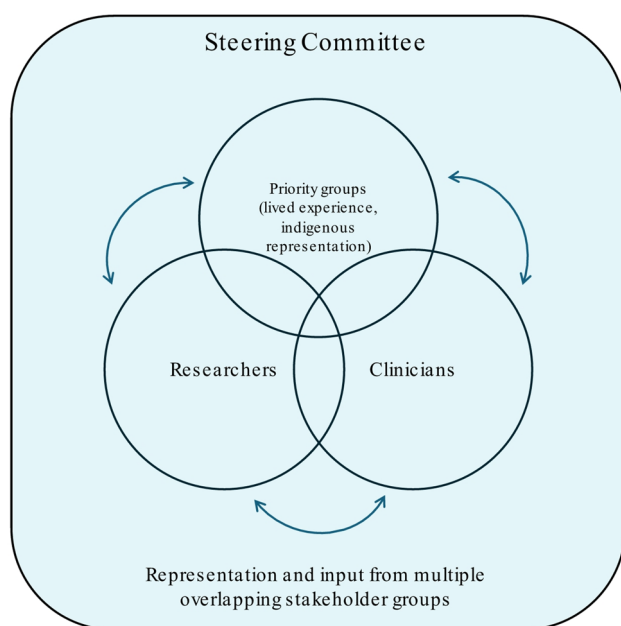


Fig. 1 Organisational structure of the steering committee, comprising representatives from key stakeholder groups

affiliated with the authors' primary research program, followed by a preliminary mapping survey and a series of initial working group meetings. The overarching aim during this early establishment stage was to review data availability and general willingness to share data amongst local networks, and to establish an initial plan that would provide a suitable scope for a pilot project.

The mapping survey was completed by 20 researchers and health clinicians who work in the HNE region. Participants were identified through existing networks and invited based on knowledge of current or prior involvement in health-related projects. The survey aimed to determine which areas of work or health research domains were the most active amongst our existing research networks and thus determine data availability for prototype development and piloting. We also sought to gauge interest in being part of a working group to guide ongoing development of the project. Of those surveyed, approximately 75% expressed a willingness to share data; and 50% expressed a willingness to be part of the working group. In terms of areas of research, ten active domains were identified across participants. Figure 2 shows the breakdown of the percentage of participants who reported the availability of data within each domain.

Based on this preliminary discovery work and a series of subsequent working group meetings with stakeholders, it was decided that the initial prototype should focus on the *mood and mental health* domain (incorporating subdomains *mood*, *psychological distress*, *"happiness"*,

social and emotional wellbeing) due to the immediate availability of data; with an aim to expand to additional domains for future iterations of the data platform. By limiting our focus to the mental health domain, we sought to ascertain suitable harmonisation algorithms, demonstrate feasibility for harmonisation and integration of data into the initial platform, and provide a more focused scope for evaluating the platform's usability, acceptability and scalability in the near future.

To build engagement and mitigate common challenges associated with data sharing, early working group members agreed that a steering committee should provide clinical oversight to the project, while ongoing stakeholder engagement through co-design methodologies would be crucial to the success of the data harmonisation platform.

Steering committee roles and responsibilities

The role of the steering committee was to ensure development of the platform complies with all national regulatory, security, privacy and ethical standards, and aligns with principles of the National Health and Medical Research Council (NHMRC) Open Access Policy [27] and the Local Health District (LHD) Strategic Plan [21]. Members of the working group along with participants from the mapping survey were contacted and invited to join the project steering committee, which typically comprised around 9 attendees at meetings. The steering committee worked with the core project team to establish clear data management practices for the platform

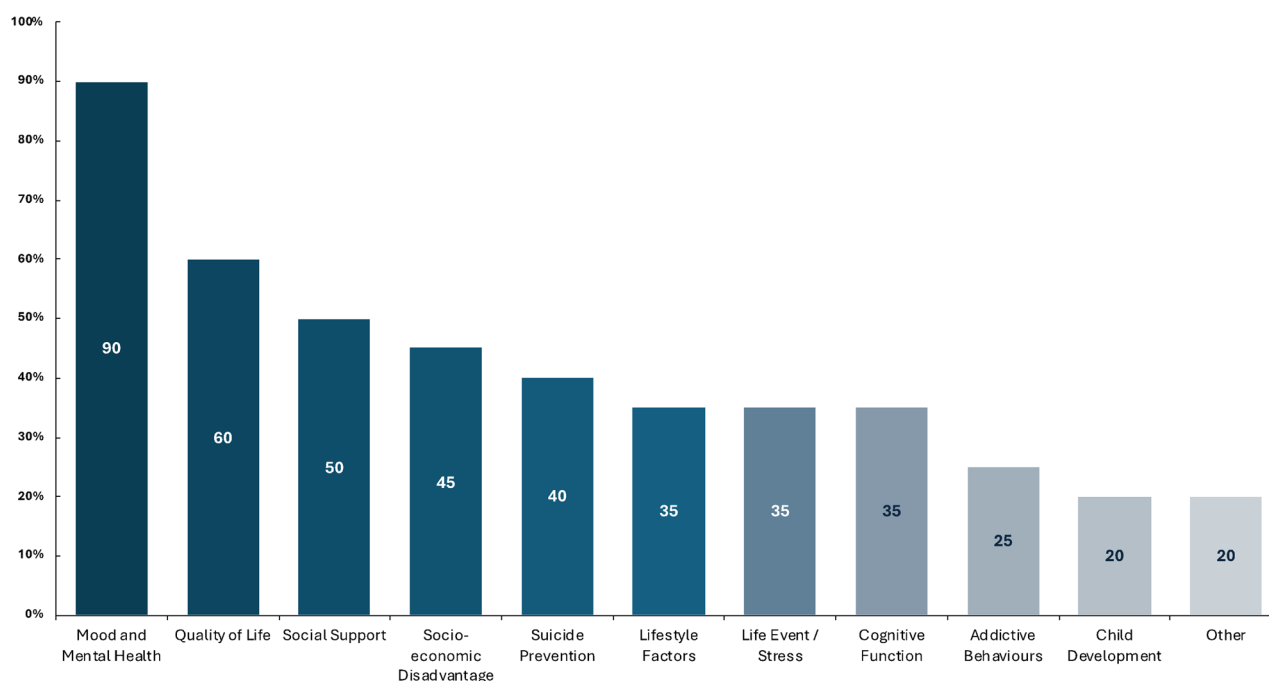


Fig. 2 Summary of participants' data availability across 10 domains of health research

and promote collaborator and stakeholder engagement, whilst providing appropriate oversight, obtaining ethical approvals as required, and ensuring processes were in place to protect data privacy and confidentiality. An early focus was development of clear protocols for data submission, transparent guidelines around data access and usage, and a data contribution plan that would facilitate easy sharing and lessen the administrative burden for project leads by allowing data to be integrated into the platform with minimal duplication of effort. Appendix A shows the data governance and consent form provided to data custodians outlining the requirements and guidelines for sharing data with the project. Data from six projects were collected during this stage, spanning community, digital mental health and substance use interventions, staff mental and wellbeing surveys, mining and creative industries, and early childhood. To be eligible for inclusion, datasets were required to include at least one routine mental health outcome measure such as the Depression, Anxiety and Stress Scale (DASS 21; [25] or the Kessler 10 [22].

Data capture system

The first step in developing the data prototype was to establish a secure, centralised repository for all datasets. REDCap [18, 19] was selected as the data management platform for the project due to its robust user rights controls, comprehensive security features, audit trails, and global uptake in health research (Over 8,000 institutions use REDCap at the time of writing). REDCap provided a scalable solution for storing de-identified data and managing variable dictionaries. Importantly, REDCap supports structured data entry. The repository was set-up so that each form can be extracted and used to collect data using the now standardised variable names and coding conventions. This approach not only facilitated integration of existing datasets but has also laid the groundwork for future studies to adopt standardised forms, promoting consistency in data collection and improving harmonisation over time.

Data harmonisation -mapping of variables

Raw de-identified data was collected and imported into the RedCap platform for cleaning and reviewing by PS, EM, and SG. A report was produced on the basis of all variables across the six datasets. Feedback was sought from the Steering Committee on variables in the report. Table 1 below shows the process involved mapping all demographic and outcome measures across datasets, identifying inconsistencies in naming conventions, coding schemes, and measurement scales. Feedback from the Steering Committee emphasised the need for clinically meaningful outputs that would be relevant to end-users, including clinicians and service planners. Moving

forward as part of the ongoing governance and oversight of the harmonised datasets, consideration of data sampling and time frame will be taken when including additional data, to mitigate the risk of overlap.

Demographic Variables: Demographic variables were harmonised to align with Australian Bureau of Statistics (ABS) standards wherever possible. Age was recorded across all six datasets; however, one dataset (Child Behaviour Checklist [1] reported age only within a narrow range (2–5 years), while Shade Plus captured older adults (54–87 years). Half the datasets recorded age as a continuous variable, with others providing categorical groupings. Gender was inconsistently coded across the datasets and overlapped with ABS standards on the collection of Sex (at birth) in one case. For example, some studies captured gender as four categories: (male, female, non-binary & other), whereas other studies used only two categories (male, female). Our composite measure consisted of three categories (male, female, other).

Country of birth was only recorded in Shade Plus. Ethnicity was collected in two datasets, with different coding in each. Indicators of Aboriginal and/or Torres Strait Islander status were present in most datasets but missing from CBCL. Coding conventions varied, making comparisons difficult to make (e.g., separate categories versus combined “Aboriginal and/or Torres Strait Islander” responses). Culturally and Linguistically Diverse (CALD) status was inconsistently applied, appearing in COVID-19 and Small Business but not in other datasets. Socio-economic indicators such as marital status, education level, and income were sparsely collected, with Shade Plus providing the most detailed breakdown of education categories, while other datasets either omitted these variables or recorded them in restricted formats.

These inconsistencies required a multi-step harmonisation process, including recoding categorical variables, standardising terminology, and applying ABS classifications for demographic fields. Where variables were missing or partially reported, composite indicators were derived where feasible. For constructs such as gender and cultural identity, harmonisation involved balancing inclusivity with comparability, ensuring that aggregated outputs remained meaningful for both research and clinical decision-making. The variability in socioeconomic measures limited the scope for pooled analysis in these domains, highlighting the need for future standardisation of data collection protocols across projects.

Outcome Variables: Outcome measures were grouped into four primary domains: mood and psychological distress, quality of life, social functioning, and alcohol and drug use. Coverage across datasets was highly variable.

Mood measures were the most consistently represented, with instruments such as the Kessler Psychological Distress Scale (K6/K10) [22], Depression Anxiety

Table 1 Demographic and outcome variables across studies

Demographics	Child Checklist	Behaviour	COVID-19	Shade Plus	Mining	Creative Industries	Small Business
Age	2-5			54-87			
Age (Categories)							
Gender							
Country of Birth							
Ethnicity							
Aboriginal and/or Torres Strait Islander							
Culturally and Linguistically Diverse							
Marital Status							
Education							
Income							
Mood Questionnaire							
Geriatric Depression Scale							
Kessler Psychological Distress Scale (K6/10)							
Anxiety/Depression Score							
Depression Anxiety Stress scale 21(DASS 21)							
Rate mental health							
Quality of Life							
Rate physical health							
Satisfaction With Life Scale (SWLS)							
Suicide Ideation Attributes Scale(SIDAS)							
WHOQOL-OLD							
Social questionnaires							
Work related questions							
Social Network Index (SNI)							
Multiple group membership							
UCLA Loneliness scale							
Alcohol/Drug use							
Alcohol use disorder identification test (AUDIT-10)							
Short Michigan Alcohol Test (SMAST)							
Drug use questions							
MH medications							

	= included
	= not included

Stress Scales (DASS-21) [25], and self-rated mental health appearing in multiple datasets. Clinical cut-points were available for all mood measures.

Quality of life measures were less consistently collected. While Shade Plus included the World Health Organisation Quality of Life measure for older people [29] measuring six quality of life facets (sensory ability, death and dying; autonomy; social participation; intimacy; past, present and future activities) and the five-item Suicide Ideation Attributes Scale (SIDAS) – measuring presence of suicidal thoughts and their severity [37]. Other datasets relied on single-item ratings of physical or mental health. The Satisfaction With Life Scale (SWLS) – a five-item scale measuring global life satisfaction [13] was

collected in the Creative Industries dataset, the codes collected originally did not match the coding of the validated tool and therefore were not included in the repository.

Work related questions were collected in all but one dataset, but the exact questions and coding differed considerably. Alcohol and drug use measures were present in over half the available datasets. The AUDIT-10 [34] was used in 50% of the available datasets. Half the datasets also asked about medication use.

Harmonisation methods will be chosen in consultation with evidence from the Maelstrom Research group [14, 39] for addressing heterogeneity between variables. The selection of harmonisation methodologies will be

dependent on the format and availability of variables in the original data and clinical and service use.

Prototype development using iterative co-design

Development of the Power BI visualisation dashboard will follow an iterative cycle, where members of the Steering Committee and individuals from broader networks representing key stakeholders (e.g., lived experience representatives) will be provided opportunities to review the prototype and provide feedback during development (using questions adapted from the Mobile Application Rating scale (uMARS) [35] on a range of design and functionality features, including but not limited to: function (performance, ease of use, navigation), aesthetics (layout, visualisations), Information (data fidelity, quality, credibility) and participants subjective appeal (would they recommend, would they use, star rating). The data science team will review and action feedback until the Steering Committee reached agreement that the dashboard will be suitable for an initial release, with required features either incorporated into the app or moved to the roadmap for a future release.

The key motivation for developing the data harmonisation platform was to enable new research and service planning questions that cannot be addressed using individual datasets or studies in isolation. By integrating data from a range of multiple diverse service and research sources, particularly within the focused yet diverse geographical region (consisting of both urban and rural/remote settings), it may be possible to explore how mental health outcomes vary within different occupation groups, proximity to and engagement with health services, community response to adversity (e.g. environmental), response to localised interventions, or temporal changes in community mental health. From a service planning perspective, collated evidence may help to identify regions or populations (e.g. workforces) with higher distress, but lower service engagement, which could then be used to inform targeted intervention and outreach initiatives, ultimately resulting in more evidence informed practice.

Discussion

The sharing and reuse of research data is central to contemporary principles of Open Science and is now considered to be a national priority [26]. In parallel with an increased emphasis on publishing in open access journals, there has inevitably arisen a greater need for data initiatives that provide new opportunities for collaborations and reuse of data. A key goal for this project was to develop a shared understanding of the new data harmonisation platform from the perspective of our key user groups and identify barriers and enablers around data sharing in the context of a regional area of Australia.

The current project focused on establishing a robust governance framework, developing streamlined data sharing processes and developing a digital platform that will facilitate data sharing and accessibility among stakeholders. Using co-design approaches, we are able to iteratively build and pilot the new data harmonisation platform and getting feedback. So far, our informal preliminary feedback from the steering committee around feasibility, applicability and potential scalability has been positive.

The first iteration of our harmonisation platform will serve as a testing ground for development of a scalable integration and harmonisation model that can be applied to current data as well as new data as it becomes available in future. As other data harmonisation initiatives have demonstrated (e.g. [10].), there is also considerable potential for regionally focused shared data platforms to foster new local research partnerships, support decision making processes, and identify gaps regarding service delivery and interventions. Additionally, access to broader, more heterogeneous datasets will allow researchers and clinicians to better explore and understand the social, emotional and economic determinants of health that are locally scoped to the unique needs of the HNE region. Future iterations will expand to other domains shown in Fig. 1, including social functioning, cognition, and health across the lifespan.

One of the major barriers encountered early in the development phase of the platform were a range of governance and regulatory challenges that required careful consideration. For instance, data were often originally collected under separate ethics approval with varying consent processes regarding the secondary use of data, with only few specifically stating that data could be shared beyond traditional methods of data dissemination. To mitigate this concern, we (1) developed guidance protocols for all prospective research to include clear statements regarding the secondary use of data; (2) liaised with ethics for guidance on the use of all retrospective data, which depending on the project status involved either variations to protocol, or to seek waiver of consent; and (3) only fully de-identified data were included in the platform – with strategies to minimise any risk to individual participant privacy or potential for re-identification (e.g. suppression rules limiting data visualisation of those with smaller sample sizes).

Implementation and future directions

Following the development of the initial prototype with a focus on mental health data, future iterations of the platform will expand to incorporate data from other domains such as social functioning, cognition, and child development. A key strength of the platform will be its capacity to promote standardisation in data collection,

leveraging REDCap's structured forms to ensure consistent variable naming, coding conventions, and recommended measures across studies. This will enable more efficient harmonisation and comparability of data over time. Additionally, we will look to understand the potential for regional scaling, including strategies for broader adoption and collaboration, and the extensibility of the platform to other regions. Future applications will be considered in terms of reporting, secondary analysis, and health systems planning.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12913-026-14552-9>.

Supplementary Material 1

Author contributions

AC and JD wrote the manuscript. PS & EP prepared the table. All authors have reviewed the manuscript.

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Data availability

Data will be available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

Our research project accepted de-identified mental health for the purposes of harmonising the datasets across the studies. It is important to note that the data custodians and chief investigators from the primary studies hold responsibility for obtaining consent from study participants, and only accepted data into the project once we have verified that participants in the primary study have consented to their data being shared for ongoing research. Our research got ethics approval was for accepting de-identified datasets for purposes of harmonisation and was conducted in accordance with the declaration of Helsinki and was approved by the Hunter New England Human Research Ethics Committee of Hunter New England Local Health District, **Reference 2024/ETH01340**.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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